Deep learning algorithm

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Gulshan V, Peng L, Coram M, et al: Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. JAMA. 2016; 316(22):2402–10.

Comment

Diabetic retinopathy (DR) is a leading cause of blindness in the working age population worldwide. In our retina service, we currently see patients who are referred by primary care physicians, endocrinologists, and comprehensive ophthalmologists if they exhibit features consistent with DR. With the growing burden of diabetes worldwide, the number of patients need to be screened for referable diabetic retinopathy (RDR) grows larger every year and is quickly overwhelming retina clinics nationwide. This predicament necessitates a comprehensive approach for screening of RDR that is quicker and more efficient than the traditional manual fundus examinations and grading of fundus photographs that we currently rely on. Gulshan and colleagues at Google propose a novel solution using deep learning technology in which machines learn how to detect patterns in data for automated detection and classification of RDR and diabetic macular edema. This technology, called backpropagation, circumvents the need to "teach" an algorithm how to identify desired features and instead learns the most predictive features directly from images previously graded by ophthalmologists. It accomplishes this by indicating how a machine should change its internal parameters to best predict the desired output of an image by creating a prediction model based on thousands of images.[2]

This automated deep learning algorithm has limitless applications, enabling fast and accurate detection of RDR while utilizing minimal manpower compared to manual grading of individual images by ophthalmologists. We now have the potential to facilitate comprehensive RDR screening that will not be limited by the large number of patients who require close follow-up. With further improvements in the technology, machines will eventually have the ability to detect subtle disease beyond the capability of any human eye. From this technology, we can learn to identify subtleties and certain disease features that may be inherently present in certain pathologies that we previously had not considered, enabling ophthalmologists to become better diagnosticians. With the endless possibilities of implementation of deep learning algorithms, it is important to consider the appropriate setting for such screening, whether it should include all individuals with metabolic syndrome or other risk factors for diabetes in the primary care setting, or individuals who have been referred to an ophthalmologist in order to "triage" the appropriate retinal care for these patients.

This technology comes with limitations, however, such as the study's complicated definition of a gold standard, limited data on "sight-threatening diabetic retinopathy," lack of data for detection of other important pathology such as glaucoma or age-related macular degeneration, and lack of data on the performance of this algorithm in populations with higher rates of diabetes.[6] Despite these limitations, this technology will alleviate some of the burden on ophthalmologists by eliminating patients without disease who do not need to be manually examined. Since this technology is relatively affordable and enables the ability to process almost 260 million images per day,[1] we must anticipate an increase in number of patients that retina specialists will need to follow-up and treat as a result of improved detection of RDR. Retina clinics are already overwhelmed by the sheer number of patients who have already been diagnosed with DR and require close follow-up for progression of disease. The identification of potentially thousand more patients each year may make retina specialist care less accessible, with increasing difficulty obtaining an appointment and longer clinic wait times. This can potentially lead to catastrophic loss of vision in patients who require urgent management of significant disease. This problem will be compounded by the inevitable false positives that will arise with large-scale implementation of such a screening tool.

These advancements in technology may ultimately facilitate the implementation of universal screening for RDR in the primary care setting. Since this technology can be utilized remotely, we will be able to easily establish a tele-ophthalmology platform that will have the potential to improve diagnosis of RDR worldwide. Haddock and colleagues have created an affordable method of obtaining fundus photographs using only a smartphone and 20D lens.[3] Marrying an easily accessible and affordable method for obtaining fundus photographs along with automated detection of RDR affords a considerable advancement in RDR screening of this caliber,